Lambda > Functions > Create function

Create function Info

Choose one of the following options to create your function.

- (●) **Author from scratch**
  Start with a simple Hello World example.
- ( ) **Use a blueprint**
  Build a Lambda application from sample code and configuration presets for common use cases.
- ( ) **Container image**
  Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

CloudFunFacts

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime | Info

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.13

Architecture | Info

Choose the instruction set architecture you want for your function code.

- ( ) arm64
- (●) x86_64

Permissions Info

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ Change default execution role

Execution role

Choose a role that defines the permissions of your function. To create a custom role, go to the IAM console.

- (●) Create a new role with basic Lambda permissions
- ( ) Use an existing role
- ( ) Create a new role from AWS policy templates

ⓘ Role creation might take a few minutes. Do not delete the role or edit the trust or permissions policies in this role.

Lambda will create an execution role named CloudFunFacts-role-b638cbrq, with permission to upload logs to Amazon CloudWatch Logs.

▶ Additional configurations

Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

Cancel | Create function

Lambda > Functions > CloudFunFacts

Successfully updated the function **CloudFunFacts**.

Code source Info

Open in Visual Studio Code

CloudFunFacts

EXPLORER

CLOUDFUNFACTS

lambda_function.py

DEPLOY

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [SELECTED: TEST1]

lambda_function.py

```python
import random
import json

def lambda_handler(event, context):
    facts = [
        "AWS S3 was launched in 2006 and still rules cloud storage.",
        "Cloud computing can save companies up to 30% on IT costs.",
        "EC2 was one of the first AWS services to change IT forever.",
        "AWS offers more than 200 services — that's more than Starbucks drinks!",
        "Cloud lets you pay-as-you-go, just like your Netflix subscription.",
        "The name 'Amazon Web Services' was first used back in 2002.",
        "AWS data centers are so secure they require palm scanners.",
        "Netflix runs most of its infrastructure on AWS.",
        "Amazon DynamoDB can handle more than 10 trillion requests per day.",
        "AWS Lambda was launched in 2014 and started the serverless trend.",
        "Cloud reduces CO₂ emissions by optimizing energy usage.",
        "AWS regions have multiple Availability Zones for reliability.",
        "Amazon originally created S3 to solve its own scaling issues.",
```

API Gateway > APIs > Create API > Create HTTP API

tep 2 - *optional*
Configure routes

tep 3 - *optional*
Define stages

tep 4
Review and create

API details

API name
An HTTP API must have a name. The name is a non-unique value you use to identify and organize your APIs. To programmatically refer to this API, use the API ID that API Gateway generates for you.

FunfactsAPI

IP address type | Info
Select the type of IP addresses that can invoke the default endpoint for your API. You don't need to redeploy your API for the update to take effect.

- (●) IPv4
  Includes only IPv4 addresses.
- ( ) Dualstack
  Includes IPv4 and IPv6 addresses.

Integrations (1) Info

Specify the backend services that your API will communicate with. These are called integrations. For a Lambda integration, API Gateway invokes the Lambda function and resp
with the response from the function. For an HTTP integration, API Gateway sends the request to the URL that you specify and returns the response from the URL.

Lambda

Remove

AWS Region	Lambda function	Version	Learn more
us-east-1	arn:aws:lambda:us-east-1:252366102382:function:CloudFunFa	2.0	

Add integration

Configure routes - *optional*

Configure routes Info

API Gateway uses routes to expose integrations to consumers of your API. Routes for HTTP APIs consist of two parts: an HTTP method and a resource path (e.g., GET /pets). You can define specific HTTP methods for your integration (GET, POST, PUT, PATCH, HEAD, OPTIONS, and DELETE) or use the ANY method to match all methods that you haven't defined on a given resource.

Method	Resource path		Integration target	
GET	/funfact	→	CloudFunFacts	Remove

Add route

Cancel | Review and create | Previous | Next

API Gateway > APIs > FunfactsAPI (0aaet190yd) > Stages

API: FunfactsAPI...(0aaet190yd)

- **Develop**
  - Routes
  - Authorization
  - Integrations
  - CORS
  - Reimport
  - Export
- **Deploy**
  - Stages
- **Monitor**
  - Metrics

Successfully created API FunfactsAPI (0aaet190yd).

Stages

Stage: $default ▼ | Deploy

Stages for FunfactsAPI

Create

Search

$default

Stage details

Delete | Edit

Details

Name	Created	Last updated
$default	August 30, 2025 5:29 PM	August 30, 2025 5:29 PM

Invoke URL

https://0aaet190yd.execute-api.us-east-1.amazonaws.com

Description

None

Attached deployment

aws | Search [Alt+S] | United States (N. Virginia) | Account ID: 2523-6610-2382

DynamoDB > Tables > Create table

Create table

Table details Info

DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

Table name

This will be used to identify your table.

CloudFacts

Between 3 and 255 characters, containing only letters, numbers, underscores (_), hyphens (-), and periods (.).

Partition key

The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.

FactID | String

1 to 255 characters and case sensitive.

Sort key - *optional*

You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.

Enter the sort key name | String

1 to 255 characters and case sensitive.

Table: CloudFacts - Items returned (15)

Actions ▼ Create item

Scan started on August 30, 2025, 18:24:31

< 1 >

FactID *(String)*	FactText
15	More data is stored in the cloud today than in all personal computers combined.
14	AWS Lambda can automatically scale from zero to thousands of requests per second.
13	The word 'cloud' became popular in the 1990s as a metaphor for the internet.
12	The fastest-growing AWS service is Amazon SageMaker, used for Machine Learning.
11	Amazon once accidentally took down parts of the internet when S3 had an outage in 2017.
10	Each AWS Availability Zone has at least one independent power source and cooling system.
9	Serverless doesn't mean there are no servers, it just means you don't manage them.
8	AWS data centers are so secure that they require retina scans and 24/7 guards.
7	More than 90% of Fortune 100 companies use AWS.
6	The 'cloud' doesn't float in the sky, it's made of giant data centers on Earth.
2	Netflix runs almost all of its infrastructure on AWS.
1	AWS S3 was one of the very first AWS services, launched in 2006.
5	AWS has more than 200 fully featured services today.
4	NASA uses AWS to store and share Mars mission data with the public.
3	Cloud computing can save companies up to 30% on IT costs.

IAM > Roles > CloudFunFacts-role-b638cbrq > Add permissions

Attach policy to CloudFunFacts-role-b638cbrq

▶ **Current permissions policies** (2)

Other permissions policies (1/1094)

Filter by Type

AmazonDynamoDBReadOnlyAccess | All types | 1 match | 1

Policy name	Type	Description
AmazonDynamoDBReadOnlyAccess	AWS managed	

Cancel | Add permissions

Lambda > Functions > CloudFunFacts

EXPLORER

CLOUDFUNFACTS

lambda_function.py

DEPLOY [UNDEPLOYED CHANGES]

You have undeployed changes.

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [SELECTED: TESTEVENT]

Create new test event

Private saved events

TestEvent

Shareable saved events

test1

ENVIRONMENT VARIABLES

lambda_function.py

```python
import boto3
import random
import json

# Create DynamoDB client
dynamodb = boto3.resource("dynamodb")
table = dynamodb.Table("CloudFacts")

def lambda_handler(event, context):
    # Scan entire table (not efficient for huge tables, but fine here)
    response = table.scan()
    items = response.get("Items", [])

    if not items:
        return {
            "statusCode": 500,
            "body": json.dumps({"error": "No facts found"})
        }

    # Pick random fact
    fact = random.choice(items)["FactText"]

    return {
        "statusCode": 200,
        "headers": {"Content-Type": "application/json"},
```

Create new test event

Invoke

Save

Event Name

MyEventName

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings

Private

This event is only available in the Lambda Console and to the event creator. You can configure a total of ten.

Deploying code

Successfully updated the function CloudFunFacts.

Amazon Bedrock > Model access

Custom models
Prompt router models
Imported models
Marketplace model deployment

Build
Agents
Flows
Knowledge Bases
Automated Reasoning
Guardrails
Prompt Management
Data Automation

Assess
Evaluations

Configure and learn
Settings
Model access
User guide

▶ Amazon (15)	0/15 access granted		
▼ Anthropic (10)	1/10 access granted		
Claude Sonnet 4 Cross-region inference	Available to request	Text	EULA
Claude 3 Sonnet	Available to request	Text & Vision	EULA
Claude 3 Opus Cross-region inference	Unavailable	Text & Vision	EULA
Claude 3 Haiku	Available to request	Text & Vision	EULA
Claude 3.5 Sonnet	Access granted	Text & Vision	EULA
Claude 3.5 Haiku Cross-region inference	Available to request	Text	EULA
Claude 3.7 Sonnet Cross-region inference	Available to request	Text & Vision	EULA
Claude 3.5 Sonnet v2 Cross-region inference	Available to request	Text & Vision	EULA
Claude Opus 4 Cross-region inference	Available to request	Text & Vision	EULA
Claude Opus 4.1 Cross-region inference	Available to request	Text & Vision	EULA
▼ Cohere (4)	0/4 access granted		

IAM > Roles > CloudFunFacts-role-b638cbrq > Add permissions

Attach policy to CloudFunFacts-role-b638cbrq

Current permissions policies (3)

Policy name	Type	Attached entities
AmazonDynamoDBRe...	AWS managed	1
AWSLambdaBasicExecuti...	Customer man...	1
AWSLambdaVPCAccessE...	Customer man...	1

Other permissions policies (1/1093)

Filter by Type

AmazonBedrockFullAccess | All types | 1 match | 1

Policy name	Type	Description
AmazonBedrockFullAccess	AWS managed	Provides full access to Amazon Bedroc...

Cancel | Add permissions

Lambda > Functions > CloudFunFacts > Edit basic settings

Ephemeral storage Info

You can configure up to 10 GB of ephemeral storage (/tmp) for your function. View pricing

512 MB

Set ephemeral storage (/tmp) to between 512 MB and 10240 MB.

SnapStart Info

Reduce startup time by having Lambda cache a snapshot of your function after the function has initialized. To evaluate whether your function code is resilient to snapshot operations, review the SnapStart c
runtimes, view pricing.

None

Supported runtimes: .NET 8 (C#/F#/PowerShell), Java 11, Java 17, Java 21, Python 3.12, Python 3.13.

Timeout

0 min 30 sec

Execution role

Choose a role that defines the permissions of your function. To create a custom role, go to the IAM console.

- (●) Use an existing role
- ( ) Create a new role from AWS policy templates

Existing role

Choose an existing role that you've created to be used with this Lambda function. The role must have permission to upload logs to Amazon CloudWatch Logs.

service-role/CloudFunFacts-role-b638cbrq

View the CloudFunFacts-role-b638cbrq role on the IAM console.

Lambda > Functions > CloudFunFacts > Edit basic settings

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service-role/CloudFunFacts-role-b638cbrq

View the CloudFunFacts-role-b638cbrq role on the IAM console.

eate new app

Give feedback Support Docs

Amplify provides a fully-managed web hosting experience and a backend building service to build fullstack apps. If you need a starter project, please visit the docs.

Deploy your app

To deploy an app from a Git provider, select one of the options below:

GitHub | BitBucket | CodeCommit | GitLab

Amplify requires read-only access to your repository.

To manually deploy an app from Amazon S3 or a Zip file, select "Deploy without Git"

Deploy without Git

Start with a template

Cancel Previous Next

Start a manual deployment

Manually upload objects to deploy your app. You can choose to drag and drop the artifacts directly, pull a zip from an existing S3 bucket or any other URL.

App name

cloud-fun-facts-generator

Branch name

production

Method

- (selected) Drag and drop
- Amazon S3
- Any URL

Zip the contents of your build output, not the top level folder

Make sure you zip the contents of your build output and not the top level folder. For example, if your build output generates a folder named "build" or "public", first navigate into that folder, select all of the contents, and zip it from there.

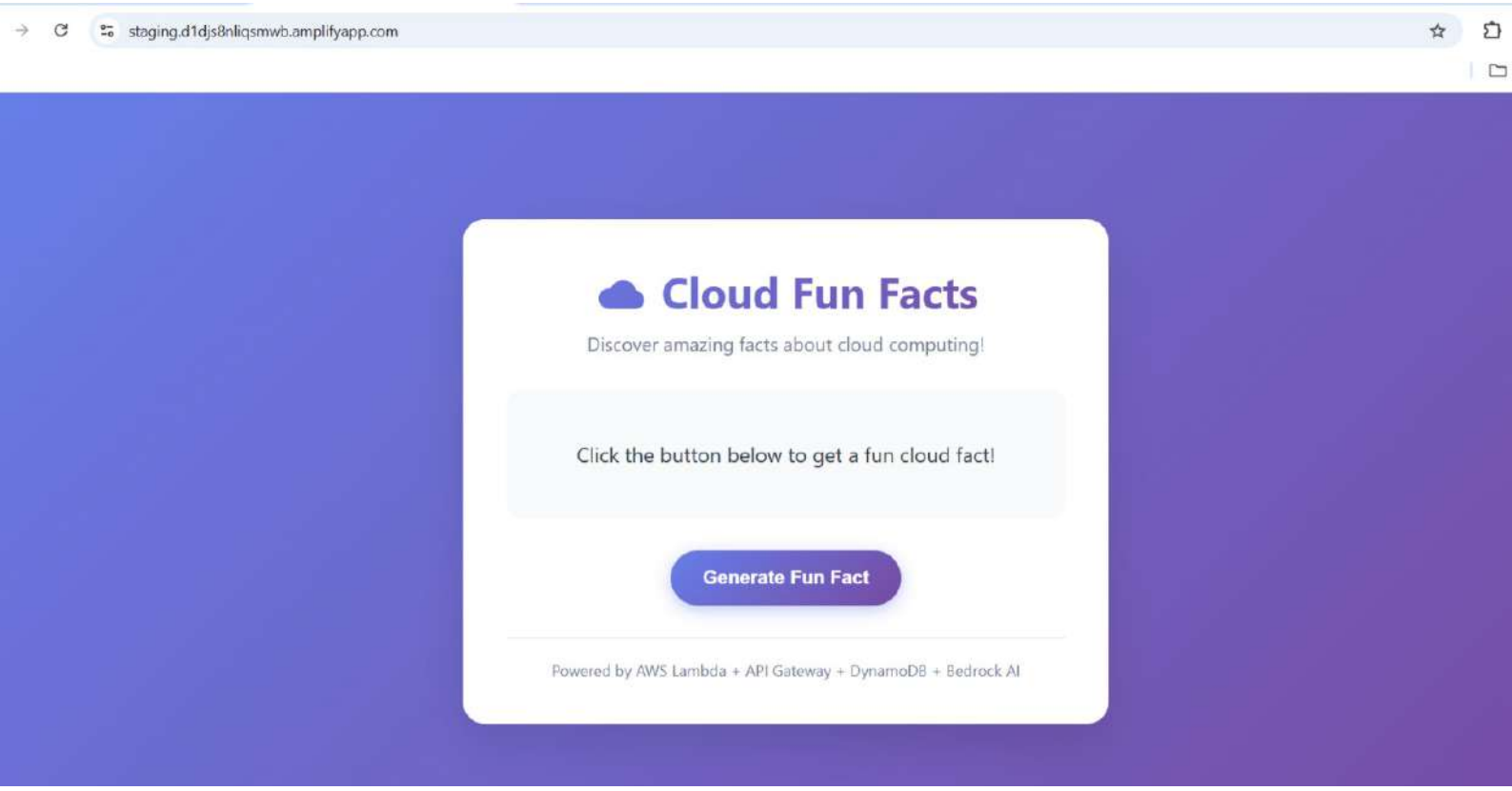
Read more

index.zip

Uploaded

Remove

Cancel | Previous | Save and deploy


staging.d1djs8nliqsmwb.amplifyapp.com
Cloud Fun Facts
Discover amazing facts about cloud computing!
Click the button below to get a fun cloud fact!
Generate Fun Fact
Powered by AWS Lambda + API Gateway + DynamoDB + Bedrock AI